MILITARY SPECIFICATION

PRIMER, ELECTRIC, XM125
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. - This specification provides the requirements and product assurance provisions for loading, assembling and packing of Primer, Electric, XM125 for use in Cartridge, APFSDS-T, XM829.

2 APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. - Unless otherwise specified, the following specifications, standards and handbooks of the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

BENEFICIAL COMMENTS (RECOMMENDATIONS, ADDITIONS, DELETIONS) AND ANY PERTINENT DATA WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT SHOULD BE ADDRESSED TO: COMMANDER, US ARMY ARMAMENT MUNITIONS AND CHEMICAL COMMAND, ATTN: AMSMC-QA, DOVER, NEW JERSEY 07801 BY USING THE SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARING AT THE END OF THIS DOCUMENT OR BY LETTER.

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-223	Powder, Black
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Table for Inspection by Attribute
MIL-STD-286	Propellants: Sampling, Inspection and Testing
MIL-STD-1234	Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1235	Single and Multiple Continuous Sampling Procedures and Tables for Inspection by Attributes

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2.1.2 Other Government documents, drawings and publications. - The following Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT MUNITIONS AND CHEMICAL COMMAND (AMCCOM)

12525016	Primer, Electric, XM125
12527222	Carton, Packing, Ammunition, for Primer, Electric, XM125
12527223	Box, Packing, Ammunition, for Primer, Electric, XM125

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Order of precedence. - In the event of a conflict between the text of this specification and the reference specifications cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 Materials. - Materials shall be in accordance with applicable drawings and specifications.

3.1.1 Primer. - The loaded primer shall comply with all requirements specified on Drawing 12525016 and with all requirements specified in applicable specifications.

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3.2 Components and parts. - The components and parts shall comply with all requirements specified on applicable drawings and specifications referenced herein.

3.3 Moisture content

3.3.1 Ignition element charge composition. - The moisture content of the ignition element charge composition at the loading station at the time of loading the ignition element shall not exceed 0.10 percent.

3.3.2 Black powder. - The moisture content of the black powder at the time of loading the closing plug shall not exceed 0.50 percent.

3.3.3 Benite. - The moisture content of the benite at the loading station at the time of loading shall not exceed 1.00 percent.

3.4 Electrical resistance test. - The head loading assembly and the loaded primers shall have an electrical resistance of not less than 0.80 ohms nor more than 1.80 ohms.

3.5 Functioning/non-functioning test (Head loading assembly)

3.5.1 Functioning. - The head loading assembly shall function when a current of 1.25 amperes, maximum, is applied.

3.5.2 Non-functioning. - The head loading assembly shall not function when a current of 200 to 210 milliamperes is applied for 18 seconds.

3.6 Lacquer in flash holes. - The lacquer in primer flash holes shall comply with illustration 1, 2 or 3 of Figure 1.

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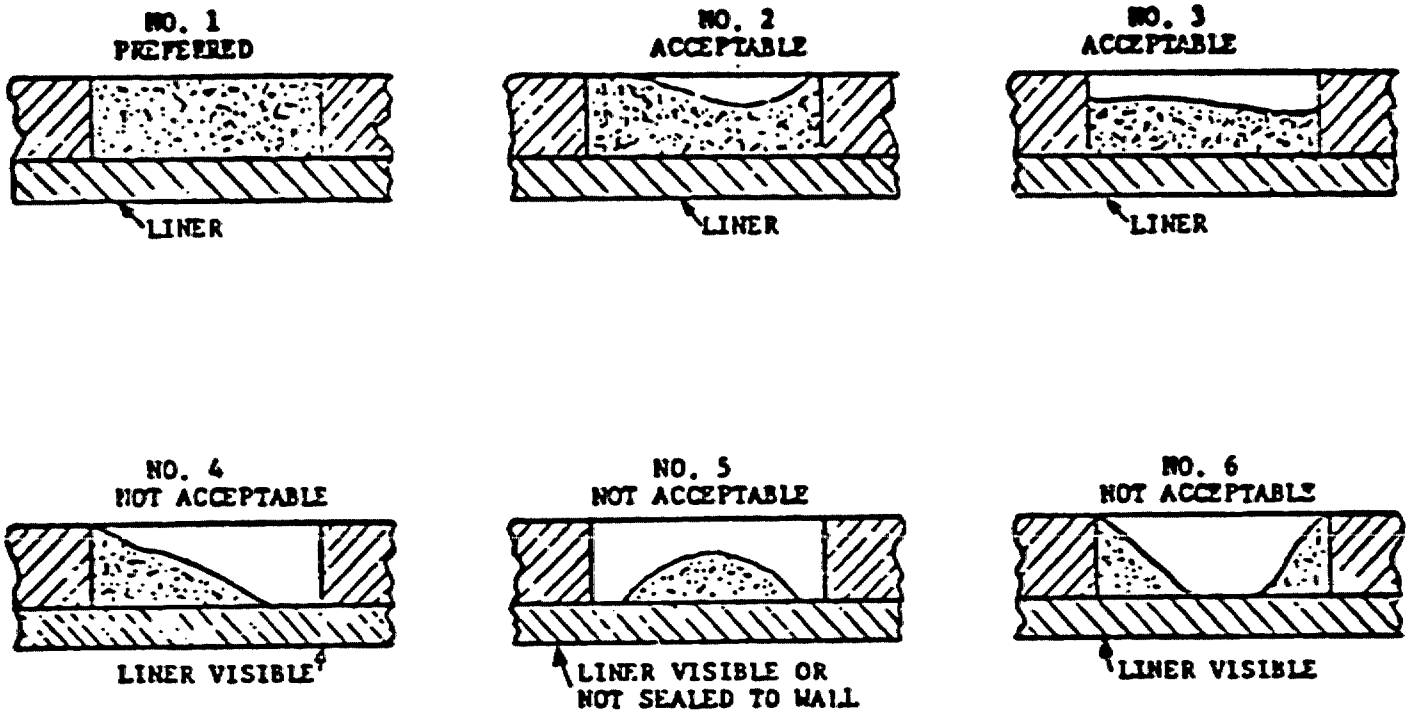


Figure 1. Primers - Standards for Lacquer Conditions in Flash Holes.

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3.7 Proving ground. - The primer shall comply with the following requirements when fired ballistically:

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.
c. There shall be no audible hangfire(s) (see 6.6.a).
d. The primer shall not squib (see 6.6.b).
e. Firing time shall not exceed 35ms (see 6.6.c).

3.8 Torque requirements

3.8.1 Assembly torque. - Threaded joints shall withstand subsequent re-application of the minimum assembly torque specified on the applicable drawing, without moving, while thread sealing compounds (if required) are wet.

3.9 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. - All parts and assemblies shall be fabricated and loaded in accordance to the applicable specifications and drawings. They shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part or assembly nor shall they be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. Required packing shall be dry.

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4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply.

4.2 Classification of inspections. - The inspection requirements specified herein are classified as follows:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample for evaluation in accordance with 4.3.2. The sample shall consist of parts which have been produced by the contractor or furnished by a supplier which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing processes, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. The first article sample shall consist of the following items in sample quantities as indicated. Whenever castings, mold or die cast parts are involved, five parts minimum from each cavity are required when there are two or more cavities.

Part/Assembly	Drawings	Quantity
Element, Ignition, Electric, Loading Assembly	8839505	10
Plug, Closing Loading Assembly	9327346	10

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Part/Assembly	Drawings	Quantity
Head Loading Assembly	12525017	400
Primer, Electric XM125	12525016	100
Carton, Packing, Ammunition for Primer, Electric, XM125	12527222	5
Box, Packing, Ammunition for Primer, Electric, XM125	12527223	5

4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. - See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE: Primer Assembly & Components		SHEET 1 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Element, ignition, electric, loading assembly (Dwg. 8839505)				
	Examination for defects	10			4.4.2.1
	Plug, closing loading assembly (Dwg. 9327346)				
	Examination for defects	10			4.4.2.8
	Head loading assembly (Dwg. 12525017)				
	Examination for defects	400			4.4.2.6 & 4.4.2.7
	Functioning 1/	200			4.5.4
	Non-functioning 1/	200			4.5.5
	Primer, electric, XM125 loading assembly (Dwg. 12525016)				
	Examination for defects 2/	50			4.4.2.9
	Proving ground 2/	50			4.5.7

NOTES:

1/ Should any sample fail to meet the applicable requirements, the First Article Sample shall be rejected.

2/ If a critical defect occurs or if 2 or more major defects occur, the First Article Sample shall be rejected.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE: Primer Assembly & Components		SHEET 2 OF 2		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Carton, packing, ammunition for primer, electric, XM125 (Dwg. 12527222)				
	Examination for defects	5			4.4.2.2 & 4.4.2.3
	Box, packing, ammunition for primer, electric, XM125 (Dwg. 12527223)				
	Examination for defects	5			4.4.2.4/4.4.2.5

NOTES

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of loaded primers shall contain:

a. Metal parts assemblies for electric ignition element from one lot interfix from one manufacturer.

b. Ignition element charge composition produced by one supplier under one continuous set of operation conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

c. Black powder from not more than one lot interfix from one supplier.

d. Primer metal parts from one lot interfix from one manufacturer.

e. Benite of one lot interfix from one manufacturer.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Element, ignition, electric, loading assembly				8839505
					NEXT HIGHER ASSEMBLY 12525017
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Charge missing or above flush		100%	3.2	Straight-edge
2	Charge composition on outside surface		100%	3.2	Visual
Major					
101	Charge insufficient		100%	3.2	Visual
102	Moisture content	1/		3.3.1	4.5.2.1

1/ One (1) sample, minimum, shall be selected from each eight (8) hours production. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.1	Element, ignition, electric, loading assembly				8839505
					NEXT HIGHER ASSEMBLY 12525017
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
103	Analysis of ignition element charge composition	2/		3.2	4.5.6
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES

2/ Five (5) grams of ignition element charge composition shall be selected from that quantity of composition used in loading a complete lot of head loading assemblies.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12527222
4.4.2.2	Carton, sealed				NEXT HIGHER ASSEMBLY 12527223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or cut		0.40%	3.2	Visual
102	Seal mislocated or incomplete		0.40%	3.2	Visual
103	Adequacy of heat seal 1/			3.2	Test
Minor					
201	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton completely		0.65%	3.2	Visual
202	Carton torn, cut or punctured		0.65%	3.2	Visual
203	Contents loose to an extent greater than permitted by packing design		0.65%	3.2	Manual
204	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.2	Visual

NOTES:

1/ Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification DOD-P-116 for the "Heat Seal Test".

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CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Carton, prior to sealing				12527222
					NEXT HIGHER ASSEMBLY 12527223
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing material missing		0.40%	3.2	Visual
102	Separator(s) damaged to the extent that primers touch each other		0.40%	3.2	Visual
Minor	None defined				

NOTE:

FIG. 685

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12527223
4.4.2.4	Box, packing, prior to sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Carton missing		0.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Box, packing sealed				12527223
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protruding nail		0.40%	3.2	Visual
102	Board broken or loose		0.40%	3.2	Visual/Manual
103	Strapping missing, broken or loose		0.40%	3.2	Visual/Manual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Strapping improperly engaged or mis-located		0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Head loading assembly, prior to assembling closing plug assembly				12525017
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Closing disc missing or damaged to the extent that explosive is exposed		0.40%	3.2	Visual
102	Retainer fails to withstand application of minimum assembly torque		0.40%	3.8.1	4.5.8.1
103	Retainer not fully seated		0.40%	3.2	Visual
104	Retainer stake missing		0.40%	3.2	Visual
105	Lacquer missing from retainer threads		0.40%	3.2	Visual
106	Electrical resistance test		100%	3.4	4.5.3
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Head loading assembly				12525017
					NEXT HIGHER ASSEMBLY 12525016
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-functioning	see 4.4.3.1.2		3.5.2	4.5.5
Major					
101	Functioning	see 4.4.3.1.1		3.5.1	4.5.4
102	Depth to electrode		0.40%	3.2	Gage
103	Flash hole obstructed		0.40%	3.2	Visual
104	Closing plug fails to withstand application of minimum assembly torque.		0.40%	3.8.1	4.5.8.1

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CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.7	Head loading assembly				12525017
					NEXT HIGHER ASSEMBLY 12525016
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Lacquer missing from closing plug threads		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

FIG. 685

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9327346
4.4.2.8	Plug, closing loading assembly				NEXT HIGHER ASSEMBLY 12525017
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Black powder charge missing		100%	3.2	Visual
102	Disc(s) missing, punctured or peeling (when applicable)		0.40%	3.2	Visual
103	Lacquer missing from disc (when applicable)		0.40%	3.2	Visual
104	Tape missing or improperly assembled (when applicable)		0.40%	3.2	Visual
105	Moisture content of black powder charge	1/	3.3.2	4.5.2.2	

1/ One (1) sample, minimum, shall be selected from each eight (8) hours production. Composite samples shall not be used. If the moisture content of the sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with the non-conforming material shall be rejected.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.9	Primer electric, XM125				12525016
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Body ruptures or becomes detached	See 4.4.3.2		3.7	4.5.7
Major					
101	Electrical resistance test		0.40%	3.4	4.5.3
102	Liner improperly positioned (prior to loading)		0.40%	3.2	Visual
103	Flash hole sealing improper (prior to loading)		0.40%	3.6	Visual
104	Weight of benite charge		0.40%	3.2	Scale
105	Length of benite charge		0.40%	3.2	Gage
106	Lacquer missing from plug threads		0.40%	3.2	Visual
107	Sealant missing from body threads		0.40%	3.2	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS AND TESTS

MIL-P-70451 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.9	Primer electric, XM125				12525016 NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
108	Body fails to withstand minimum assembly torque (prior to curing)		0.40%	3.8.1	4.5.8.2
109	Moisture content of benite 1/			3.3.3	4.5.2.3
110	Failure to fire	See 4.4.3.2		3.7	4.5.7
111	Hangfire occurs	See 4.4.3.2		3.7	4.5.7
112	Squib occurs	See 4.4.3.2		3.7	4.5.7
113	Length, total, maximum		0.40	3.2	Gage
114	Presence of Benite (prior to tip, primer assembly)		0.40	3.2	Visual
Maj:					
201	Depth to plug		0.65%	3.2	Gage
202	Firing time incorrect	See 4.4.3.2			4.5.7
203	Marking incorrect or unidentifiable		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.10	Visual

Notes

1/ One 100 gram sample of benite shall be selected at the loading station, during each 8 hours production. If the moisture content of the sample exceeds the applicable requirement the primers loaded since the last acceptance sample shall be rejected.

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4.4.3 Testing

4.4.3.1 Head loading assembly

4.4.3.1.1 Functioning

4.4.3.1.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the requirements specified, five hundred (500) samples shall be selected from each lot for test as specified in 4.5.4. If four (4) or more failures occur, the lot shall be rejected.

4.4.3.1.1.2 After three consecutive lots have complied with the criteria of 4.4.3.1.1.1, a double sampling plan in accordance with Code Letter L and an AQL of 0.25 percent shall be used for each lot.

4.4.3.1.2 Non-functioning

4.4.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the specified requirement, two hundred (200) samples shall be selected from each lot and tested in accordance with 4.5.5. If any assembly functions, the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. (Destructive test)

4.4.3.1.2.2 After three consecutive lots have complied with the criteria of 4.4.3.1.2.1 eighty head loading assemblies shall be selected from each lot for this test. If any assembly functions the lot shall be rejected. Assemblies subjected to this test shall not be returned to the lot. (Destructive test)

4.4.3.2 Proving ground - (see 3.7). - The primer shall be observed for evidence of the following defects:

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TABLE II

Defects	Classification
Body ruptures or becomes detached	Critical
Failure to fire	Major
Assembly hangfires	Major
Assembly squibs	Major
T4 firing time excessive	Major

4.4.3.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, thirty-five (35) primers shall randomly be selected from each lot for this test. If a critical defect occurs or one (1) or more major defects occur, the lot shall be rejected.

4.4.3.2.2 After three (3) consecutive lots have met the criteria specified in 4.4.3.2.1, ten (10) primers shall be randomly selected from each lot for test. If one (1) critical defect occurs or if one (1) or more major defect occur, the lot shall be rejected.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

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4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Moisture content

4.5.2.1 Igniter charge composition. - The moisture content of the igniter charge composition shall be determined as specified in Method 101.5 of MIL-STD-1234 except that a five (5) gram sample shall be used.

4.5.2.1.1 Alternate method. - The moisture content of the igniter charge composition may be determined in accordance with the electrolytic hydrometer described in Method 101.4 of MIL-STD-286 using a sample size not greater than 0.2 grams and a temperature of 70 degrees Centigrade.

4.5.2.2 Black powder. - The moisture content of the black powder shall be determined in accordance with the procedure specified in MIL-P-223.

4.5.2.3 Benite. - The moisture content of the benite shall be tested in accordance with Method 102.1 of standard MIL-STD-286.

4.5.3 Electrical resistance. - The head loading assembly (dwg. 12525017) and the primer assembly (dwg. 12525016), shall be tested for electrical resistance by applying an electrical current of ten (10) milliamperes, maximum, using the electrode as one contact point and the head as the other.

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4.5.4 Functioning. - The head loading assembly shall be tested for electrical functioning by applying an electrical current of 1.25 amperes, maximum, using the electrode for one contact point and the head for the other.

4.5.5 Non-functioning. - The head loading assembly shall be tested for non-functioning by applying an electrical current of two hundred (200) to two hundred ten (210) milliamperes for eighteen seconds, minimum, using the electrode as one contact point and the head as the other.

4.5.6 Determination of ignition charge composition. - The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following:

Component	Percentage
Potassium Chlorate & Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2%
Charcoal	20.0 ± 1.5%

4.5.6.1 Charcoal. - Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70°C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to break up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5°C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B) \times (100)}{A}$$

A = Beginning sample weight

B = Empty crucible weight

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4.5.6.2 Lead thiocyanate. - Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (nitric acid used shall be water-white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for 5 minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.5.6.3 Potassium chlorate and gum solids. - The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent.

4.5.7 Proving ground. - The metal parts security characteristics of the primer shall be evaluated by assembling the primer into test rounds, which have a piezo equivalent pressure between 6200 Bars and 6700 Bars, and firing the round in an XM256 gun. The pressure shall be measured 25.4 cm from the breech. If any round fired has a pressure below the minimum, the lot shall be referred to the U.S. Army Armament Munitions and Chemical Command, Attn: DRSMC-QAT-A for disposition. If any round fired has a pressure above the maximum, the lot shall be accepted provided no ballistic failures have occurred. Should ballistic failures occur on a round which exceeds the maximum pressure, the lot shall be referred as above.

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4.5.8 Torque testing

4.5.8.1 Head loading assembly. - Whenever lacquer is used the torque test shall be performed while lacquer is still wet (within 5 minutes after the lacquer application). The torque shall be applied in a tightening direction, while holding the head. The joint under test shall be marked whenever possible to provide a means of detecting any movement between adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

4.5.8.2 Primer loading assembly. - Prior to curing of sealant, the torque shall be applied in a tightening direction, while holding the head. The joint under test shall be marked to provide a means of detecting any movement between adjacent parts. Any discernible movement between adjacent parts shall be regarded as a torque test failure.

5 PACKAGING

5.1 Packaging and marking

5.1.1 Level A. - The primers shall be packaged and marked in accordance with dwg. 12527222.

5.1.2 Level B. - The primers shall be packaged and marked in accordance with dwg. 12527222.

5.2 Packing and marking

5.2.1 Level A. - The packaged primers shall be packed and marked in accordance with dwg. 12527223.

5.2.2 Level B. - The packaged primers shall be packed and marked in accordance with dwg. 12527223.

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6 NOTES

6.1 Intended use. - The primer covered by this specification is intended for use in the 120mm kinetic energy (KE) cartridges.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals - See MIL-A-48078. - Submit designs as required to Commander, AMCCOM, ATTN: DRSMC-QAT-I(D), Dover, New Jersey 07801.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Munitions and Chemical Command (AMCCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, and U.S. Army Research and Development Command drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

6.5 Check test for possible deterioration of primers. - The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of XM125 Primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the XM125 is assembled, does not require the performance of the following check test, the contracting officer shall contractually provide for compliance of check test requirements. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

6.5.1 Moisture content of benite. - A sample of 25 primers shall be selected for this test. If one or more primers fail to comply with the requirement specified in 3.3.3, the lot shall be rejected. The test shall be performed in accordance with Standard MIL-STD-286, Method 102.1.

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6.5.2 Electrical resistance. - This test shall be performed 100 percent. Any primer assembly which fails to comply with the requirement specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.3.

6.5.3 Functioning (head loading assembly). - 300 primers shall be selected from each lot for this test. The head loading assemblies shall be removed from the primers and tested as specified in 4.5.5. If one or more assemblies fail to comply with the applicable requirement the lot shall be rejected.

6.6 Definitions

a. Hangfire. - A perceptible delay occurring between the instant application of energy and the functioning of primer/propellant.

b. Squib. - A low order explosion or slow rate burning as exhibited by no functioning of the propellant charge.

c. Firing time (T_4). - The time interval between application of voltage to the primer and emergence of the projectile sealing ring from the gun tube.

6.7 Submission of test data. - In addition to the normal distribution of records, when the primers are procured by the US Army Armament Munitions and Chemical Command, one copy of all ballistic test data shall be forwarded to each of the following:

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Commander
US Army Armament Munitions and Chemical Command
ATTN: DRSMC-QAT-T(D), DRSMC-LCU-CT(D) and DRSMC-LCA-G(D)
Dover, NJ 07801

Commander
AMCCOM
ATTN: DRSMC-QAD(R)
Rock Island, IL 01201

Custodian:

ARMY-AR

Preparing Activity:

ARMY-AR

Project Number: 1315-A586

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-70451	PRIMER, ELECTRIC, XM125 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD Form 1426 PREVIOUS EDITION IS OBSOLETE

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